

GCSE Mathematics

Predicted Paper 2 — Higher Tier

Set D

Pearson Edexcel style (1MA1 / 2H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

- Calculator
- Ruler graduated in centimetres and millimetres
- Protractor
- Pair of compasses
- Pen, HB pencil, eraser

Instructions

- Answer **all** 22 questions in the spaces provided.
 - You must show all your working.
- Diagrams are NOT accurately drawn unless stated otherwise.
 - Calculators may be used.
- If your calculator does not have a π button, take π to be 3.142.

Advice

- Read each question carefully before you start.
- The marks for each question are shown in brackets — use them as a time guide.
- Try to answer every question. Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

1. The price of a sofa is £640 before VAT. VAT is charged at 20%. Work out the total price of the sofa including VAT.

(2 marks)

2. Two numbers are 84 and 126.

(a) Write 84 and 126 as products of their prime factors.

(2 marks)

(b) Hence find the highest common factor (HCF) of 84 and 126.

(1 marks)

(c) Find the lowest common multiple (LCM) of 84 and 126.

(1 marks)

3. A train travels 270 km in 2 hours 15 minutes. Work out the average speed of the train in km/h.

(3 marks)

4. Solve $3(x - 4) \leq 5x + 2$ and represent your solution on a number line.

(3 marks)

5. The stem and leaf diagram shows the marks scored by 28 students in a test.

Stem	Leaf
1	2 5 8
2	0 3 3 6 7 9
3	1 2 4 5 5 7 8 8 9
4	0 1 2 4 6 8
5	1 3 5 7

Key: 1|2 means 12

(a) Write down the median mark.

(1 marks)

(b) Work out the range of the marks.

(1 marks)

(c) Work out the percentage of students who scored 35 or above.

(3 marks)

6. Solve $x^2 + 8x - 5 = 0$ by completing the square. Give your answers in the form $a \pm b\sqrt{c}$ where a, b, c are integers.

(4 marks)

7. A bag contains 5 red counters and 3 yellow counters. A counter is taken at random, the colour is noted, and the counter is replaced. This is done three times.

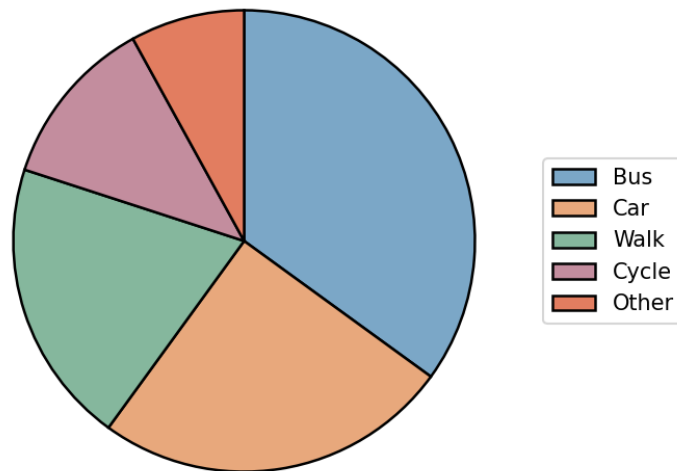
(a) Work out the probability that all three counters are red.

(2 marks)

(b) Work out the probability that exactly two of the three counters are red.

(3 marks)

8. The pie chart shows information about how 400 students travel to school.



(a) Estimate the number of students who travel by bus. (Bus represents 35% of the chart.)

(2 marks)

(b) The angle for 'Walk' is 72° . Show that this is consistent with 20% of students walking.

(2 marks)

9. Given that $f(x) = 2x - 5$, find:

(a) $f(7)$

(1 marks)

(b) The value of x for which $f(x) = 11$

(2 marks)

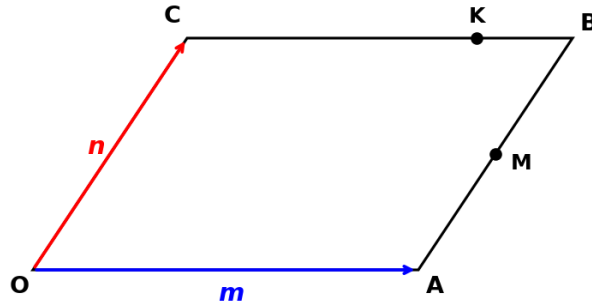
(c) $f^{-1}(x)$, the inverse function.

(2 marks)

10. Show that $(5 - \sqrt{7})^2$ can be written in the form $a - b\sqrt{7}$, where a and b are integers.

(3 marks)

11. OABC is a parallelogram. $\mathbf{OA} = \mathbf{m}$ and $\mathbf{OC} = \mathbf{n}$. M is the midpoint of AB. K is the point on BC such that $BK : KC = 1 : 3$.



(a) Express the vector $\mathbf{MK}$ in terms of $\mathbf{m}$ and $\mathbf{n}$. Simplify your answer.

(4 marks)

(b) L is the midpoint of OA. Show that MK is parallel to LC.

(2 marks)

12. A rectangular garden has length $(x + 4)$ metres and width $(x - 1)$ metres. The area of the garden is 36 m^2 .

(a) Show that $x^2 + 3x - 40 = 0$.

(2 marks)

(b) Solve the equation to find the length and width of the garden.

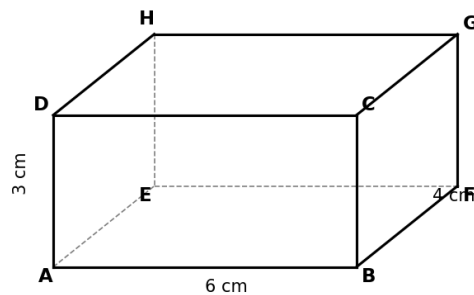
(3 marks)

13. Simplify $(4x^5)^2 \div (2x^3)^4$.

(3 marks)

14. The diagram shows cuboid ABCDEFGH with dimensions AB = 6 cm, BC = 4 cm and CG = 3 cm. Work out the length of the diagonal AG. Give your answer correct to 3 significant figures.

(4 marks)



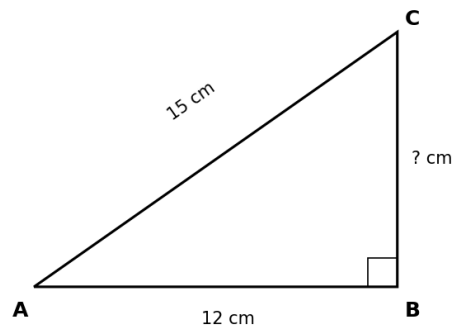
15. The value of a car decreases by 15% in the first year and by 10% in each subsequent year. A car is bought new for £20,000. Work out its value after 4 years. Give your answer to the nearest pound.

(4 marks)

16. Simplify fully $(x^2 - 4) / (x^2 + 5x + 6) \times (x + 3) / (x + 2)$.

(4 marks)

17. Triangle ABC is right-angled at B. AC = 15 cm and AB = 12 cm.



(a) Work out the length of BC.

(2 marks)

(b) Work out the size of angle BAC. Give your answer correct to 1 decimal place.

(2 marks)

18. y is directly proportional to the cube of x . When $x = 2$, $y = 56$.

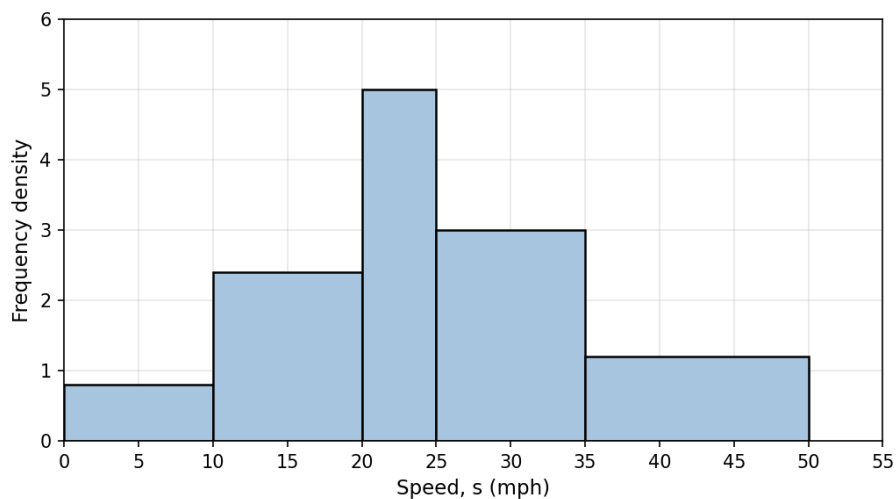
(a) Express y in terms of x .

(2 marks)

(b) Find the value of x when $y = 189$.

(3 marks)

19. The histogram shows information about the speeds of cars passing a checkpoint.



(a) Calculate an estimate of the total number of cars represented.

(2 marks)

(b) The speed limit is 30 mph. Estimate the number of cars exceeding the speed limit.

(3 marks)

20. The first five terms of a sequence are 3, 9, 19, 33, 51.

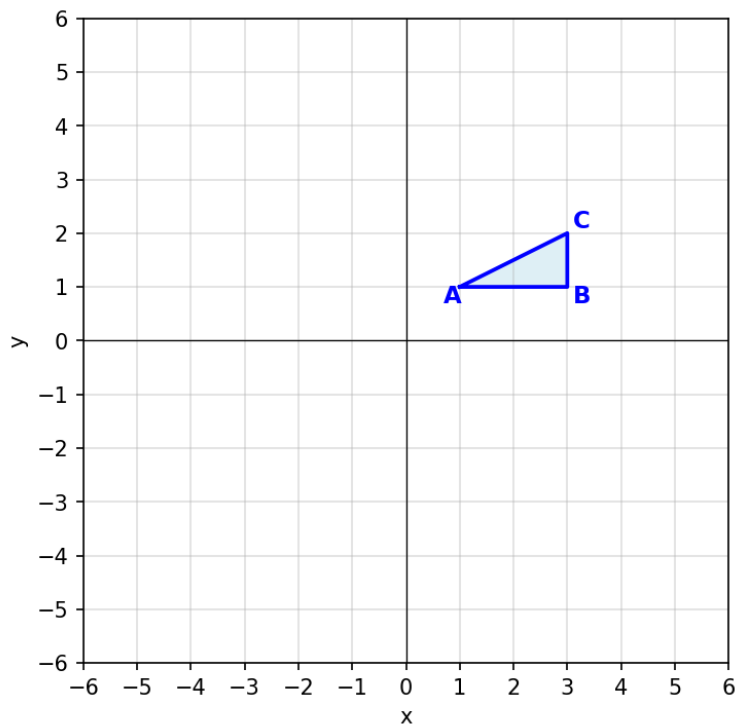
(a) Show that the n th term of the sequence is $2n^2 + 1$.

(3 marks)

(b) Work out the 12th term of the sequence.

(1 marks)

21. Triangle T has vertices A(1, 1), B(3, 1) and C(3, 2), as shown on the grid.



(a) Triangle T is reflected in the line $y = x$ to give triangle T'. Write down the coordinates of A', B' and C'.

(2 marks)

(b) Triangle T is enlarged by scale factor -2 with centre of enlargement at the origin. Write down the coordinates of the image of A under this enlargement.

(2 marks)

22. Prove algebraically that the difference between the squares of any two consecutive integers is always equal to the sum of those two integers.

(4 marks)

TOTAL FOR PAPER: 80 MARKS
END

Mark Scheme — Predicted Paper 2 Set D

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	£768	M1: 640×1.2 OR $640 + 0.2 \times 640$. A1: 768. [2]
2(a)	$84 = 2^2 \times 3 \times 7$; $126 = 2 \times 3^2 \times 7$	B1: 84 correct. B1: 126 correct. [2]
2(b)	HCF = 42	B1ft: $2 \times 3 \times 7 = 42$. [1]
2(c)	LCM = 252	B1ft: $2^2 \times 3^2 \times 7 = 252$. [1]
3	120 km/h	M1: converts 2h 15 min to 2.25 h. M1: $270 / 2.25$. A1: 120. [3]
4	$x \geq -7$	M1: $3x - 12 \leq 5x + 2$. M1: $-14 \leq 2x \Rightarrow x \geq -7$. A1: solution stated with number line (filled circle at -7 , arrow to right). [3]
5(a)	35	B1: identifies median position (14th/15th value) and reads 35. [1]
5(b)	Range = $57 - 12 = 45$	B1: 45. [1]
5(c)	57.1% (16/28)	M1: counts values ≥ 35 : 35, 35, 37, 38, 38, 39, 40, 41, 42, 44, 46, 48, 51, 53, 55, 57 = 16. M1: $16/28 \times 100$. A1: 57.1% (or 4/7). [3]
6	$x = -4 \pm \sqrt{21}$	M1: $(x + 4)^2 - 16 - 5 = 0$. M1: $(x + 4)^2 = 21$. M1: $x + 4 = \pm\sqrt{21}$. A1: $x = -4 \pm \sqrt{21}$. [4]
7(a)	125/512	M1: $(5/8)^3$. A1: 125/512. [2]
7(b)	225/512	M1: identifies $3 \times (5/8)^2 \times (3/8)$. M1: $3 \times 25/64 \times 3/8 = 225/512$. A1: 225/512. [3]
8(a)	140 students	M1: $35\% \times 400$. A1: 140. [2]
8(b)	Shown	M1: $72/360 = 0.2 = 20\%$. A1: states this confirms 20%. [2]
9(a)	9	B1: $2(7) - 5 = 9$. [1]
9(b)	$x = 8$	M1: $2x - 5 = 11$. A1: $x = 8$. [2]
9(c)	$f^{-1}(x) = (x + 5)/2$	M1: rearranges $y = 2x - 5$. A1: $(x + 5)/2$. [2]
10	$a = 32$, $b = 10$	M1: $(5 - \sqrt{7})(5 - \sqrt{7}) = 25 - 5\sqrt{7} - 5\sqrt{7} + 7$. M1: simplifies to $32 - 10\sqrt{7}$. A1: $a = 32$, $b = 10$. [3]
11(a)	$\mathbf{MK} = (3/4)\mathbf{n} - (1/2)\mathbf{m}$	M1: $\mathbf{OM} = \mathbf{m} + (1/2)\mathbf{n}$. M1: $\mathbf{OK} = \mathbf{OB} + (1/4)\mathbf{BC} = (\mathbf{m} + \mathbf{n}) + (1/4)(-\mathbf{m})$. M1: $\mathbf{OK} = (3/4)\mathbf{m} + \mathbf{n}$. A1: $\mathbf{MK} = (3/4)\mathbf{m} + \mathbf{n} - \mathbf{m} - (1/2)\mathbf{n} = -(1/4)\mathbf{m} + (1/2)\mathbf{n}$. [4]
11(b)	Shown: $\mathbf{MK} = (1/2) \times \mathbf{LC}$	M1: $\mathbf{OL} = (1/2)\mathbf{m}$, so $\mathbf{LC} = \mathbf{OC} - \mathbf{OL} = -(1/2)\mathbf{m} + \mathbf{n}$. A1: shows $\mathbf{MK} = (1/2)\mathbf{LC}$, hence parallel. [2]
12(a)	Shown	M1: $(x + 4)(x - 1) = 36$. M1: $x^2 + 3x - 4 = 36$. A1: $x^2 + 3x - 40 = 0$. [2]
12(b)	Length = 9 m, width = 4 m	M1: factorises $(x + 8)(x - 5) = 0$. M1: $x = 5$ (reject -8 as negative). A1: length = 9, width = 4. [3]
13	x^{-2} (or $1/x^2$)	M1: $16x^{10} \div 16x^{12}$. M1: simplifies. A1: x^{-2} or $1/x^2$. [3]
14	$AG = 7.81$ cm (3 s.f.)	M1: face diagonal $AC^2 = 6^2 + 4^2 = 52$. M1: $AG^2 = AC^2 + 3^2 = 61$. M1: $AG = \sqrt{61}$. A1: 7.81 (3 s.f.). [4]
15	£12,393	M1: 20000×0.85 . M1: $\times 0.90^3$. M1: 17000×0.729 . A1: 12,393. [4]
16	$(x - 2)/(x + 2)$ · correction needed — see solutions	M1: factorises numerator of first fraction: $(x + 2)(x - 2)$. M1: factorises denominator of first fraction: $(x + 2)(x + 3)$. M1: cancels and multiplies. A1: $(x - 2)$. [4]
17(a)	$BC = 9$ cm	M1: $BC^2 = 15^2 - 12^2 = 81$. A1: 9. [2]
17(b)	Angle BAC = 36.9°	M1: $\cos(\text{BAC}) = 12/15$ OR $\sin(\text{BAC}) = 9/15$. A1: 36.9° (1 d.p.). [2]

Q	Answer	Mark scheme
18(a)	$y = 7x^3$	M1: $y = kx^3$, $56 = k \times 8$. A1: $k = 7$, $y = 7x^3$. [2]
18(b)	$x = 3$	M1: $189 = 7x^3$. M1: $x^3 = 27$. A1: $x = 3$. [3]
19(a)	Total = 105 cars	M1: calculates frequencies (fd $\times$ width): 0–10: 8; 10–20: 24; 20–25: 25; 25–35: 30; 35–50: 18. M1: sums frequencies. A1: $8 + 24 + 25 + 30 + 18 = 105$. [2]
19(b)	≈ 33 cars	M1: identifies the 25–35 bar covers speeds 25 to 35; half of this bar (speeds 30–35) represents $5 \times 3.0 = 15$ cars. M1: adds full 35–50 bar = 18 cars. A1: $15 + 18 = 33$ cars. [3]
20(a)	Shown	M1: identifies first differences (6, 10, 14, 18) and second differences (4, 4, 4) $\Rightarrow$ coefficient of n^2 is 2. M1: tests: $2(1)^2 = 2$, but first term is 3, so + 1. A1: shows nth term = $2n^2 + 1$ by checking against terms. [3]
20(b)	289	B1: $2(144) + 1 = 289$. [1]
21(a)	A'(1, 1), B'(1, 3), C'(2, 3)	M1: applies reflection $(x, y) \rightarrow (y, x)$. A1: all three image points correct. [2]
21(b)	A'(-2, -2)	M1: applies scale factor -2 to (1, 1). A1: (-2, -2). [2]
22	Proof shown	M1: lets integers be n and $(n + 1)$. M1: difference of squares = $(n + 1)^2 - n^2 = 2n + 1$. M1: sum of the two integers = $n + (n + 1) = 2n + 1$. A1: clear conclusion that they are equal. [4]

Worked Solutions — Predicted Paper 2 Set D

Higher Tier · Edexcel 1MA1 style

Q1: VAT

VAT = 20% of 640 = $0.2 \times 640 = £128$. Or use the multiplier directly: $640 \times 1.20 = £768$.

Total price = **£768**.

Q2: HCF and LCM

(a) $84 = 2 \times 42 = 2 \times 2 \times 21 = 2^2 \times 3 \times 7$.

$126 = 2 \times 63 = 2 \times 9 \times 7 = 2 \times 3^2 \times 7$.

(b) HCF: take the lowest power of each common prime: $2 \times 3 \times 7 = 42$.

(c) LCM: take the highest power of each prime: $2^2 \times 3^2 \times 7 = 252$.

Q3: Average speed

Convert time to hours: 2 hours 15 min = $2 + 15/60 = 2.25$ hours.

Speed = distance / time = $270 / 2.25 = 120$ km/h.

Q4: Linear inequality

$3(x - 4) \leq 5x + 2 \Rightarrow 3x - 12 \leq 5x + 2$.

$-14 \leq 2x$, so $x \geq -7$.

Solution: $x \geq -7$. On a number line: filled circle at -7 , arrow pointing right.

Q5: Stem and leaf

The 28 values in ascending order: 12, 15, 18, 20, 23, 23, 26, 27, 29, 31, 32, 34, 35, 35, 37, 38, 38, 39, 40, 41, 42, 44, 46, 48, 51, 53, 55, 57.

(a) Median = average of 14th and 15th values = $(35 + 37)/2 = 36$.

(Accept 35 if the 14th value alone is read, or 36 if averaged.)

(b) Range = $57 - 12 = 45$.

(c) Count values ≥ 35 : from 35 onwards there are 16 students.

Percentage = $16/28 \times 100 = 57.1\%$ (to 1 d.p.).

Q6: Completing the square

$x^2 + 8x - 5 = 0$.

Complete the square: $(x + 4)^2 - 16 - 5 = 0$.

$(x + 4)^2 = 21$.

$x + 4 = \pm\sqrt{21}$.

$x = -4 \pm \sqrt{21}$ (so $a = -4$, $b = 1$, $c = 21$).

Q7: Probability with replacement

(a) $P(\text{red}) = 5/8$ each time. $P(\text{all 3 red}) = (5/8)^3 = 125/512$.

(b) 'Exactly 2 red' can happen in 3 ways: RRY, RYR, YRR.

Each has probability $(5/8)^2 \times (3/8) = 75/512$.

Total = $3 \times 75/512 = \mathbf{225/512}$.

Q8: Pie chart

(a) 35% of 400 = $0.35 \times 400 = \mathbf{140}$ students.

(b) Angle for 20% of a full circle = $0.20 \times 360 = 72^\circ$. ✓

Q9: Functions

(a) $f(7) = 2(7) - 5 = \mathbf{9}$.

(b) $2x - 5 = 11 \Rightarrow 2x = 16 \Rightarrow \mathbf{x = 8}$.

(c) Let $y = 2x - 5$. Solve for x : $x = (y + 5)/2$. So $f^{-1}(x) = \mathbf{(x + 5)/2}$.

Q10: Surds squared

$$(5 - \sqrt{7})^2 = (5 - \sqrt{7})(5 - \sqrt{7}).$$

$$= 25 - 5\sqrt{7} - 5\sqrt{7} + (\sqrt{7})^2$$

$$= 25 - 10\sqrt{7} + 7 = \mathbf{32 - 10\sqrt{7}}.$$

So $a = 32$, $b = 10$.

Q11: Vectors

(a) $\mathbf{OM} = \mathbf{m} + (1/2)\mathbf{n}$ (since $AB = OC = \mathbf{n}$).

$$\mathbf{BC} = \mathbf{OC} - \mathbf{OB} = \mathbf{n} - (\mathbf{m} + \mathbf{n}) = -\mathbf{m}.$$

$$\mathbf{OK} = \mathbf{OB} + (1/4)\mathbf{BC} = (\mathbf{m} + \mathbf{n}) + (1/4)(-\mathbf{m}) = (3/4)\mathbf{m} + \mathbf{n}.$$

$$\mathbf{MK} = \mathbf{OK} - \mathbf{OM} = (3/4)\mathbf{m} + \mathbf{n} - \mathbf{m} - (1/2)\mathbf{n} = \mathbf{-(1/4)m + (1/2)n}.$$

(b) L is the midpoint of OA, so $\mathbf{OL} = (1/2)\mathbf{m}$.

$$\mathbf{LC} = \mathbf{OC} - \mathbf{OL} = \mathbf{n} - (1/2)\mathbf{m} = \mathbf{-(1/2)m + n}.$$

$$\text{From (a), } \mathbf{MK} = \mathbf{-(1/4)m + (1/2)n} = (1/2) \times \mathbf{-(1/2)m + n} = (1/2)\mathbf{LC}.$$

Since $\mathbf{MK}$ is a scalar multiple of $\mathbf{LC}$, $\mathbf{MK}$ is parallel to $\mathbf{LC}$. ✓

Q12: Quadratic word problem

(a) Area = length $\times$ width: $(x + 4)(x - 1) = 36$.

$$\text{Expand: } x^2 + 3x - 4 = 36 \Rightarrow \mathbf{x^2 + 3x - 40 = 0.} \quad \checkmark$$

(b) Factorise: $(x + 8)(x - 5) = 0$. So $x = -8$ or $x = 5$.

Reject $x = -8$ (lengths cannot be negative). So $x = 5$.

Length = $5 + 4 = \mathbf{9 \text{ m}}$, width = $5 - 1 = \mathbf{4 \text{ m}}$.

Q13: Index laws

$$(4x^5)^2 = 16x^{10}.$$

$$(2x^3)^4 = 16x^{12}.$$

$$\text{Divide: } 16x^{10} / 16x^{12} = x^{10-12} = \mathbf{x^{-2}} \text{ (or } 1/x^2\text{)}.$$

Q14: 3D Pythagoras

Find face diagonal AC first (rectangle ABCD on bottom face):

$$AC^2 = AB^2 + BC^2 = 6^2 + 4^2 = 36 + 16 = 52.$$

Then space diagonal AG (in right triangle ACG, since CG is vertical):

$$AG^2 = AC^2 + CG^2 = 52 + 9 = 61.$$

$$AG = \sqrt{61} = \mathbf{7.81 \text{ cm (3 s.f.)}}.$$

Q15: Compound percentage change

Year 1: $20000 \times 0.85 = \text{£}17000$.

Years 2 to 4: each year multiplied by 0.90.

Value after 4 years = $17000 \times (0.90)^3 = 17000 \times 0.729 = \text{£}12393$.

£12,393 to the nearest pound.

Q16: Algebraic fractions

$$x^2 - 4 = (x + 2)(x - 2).$$

$$x^2 + 5x + 6 = (x + 2)(x + 3).$$

Substitute: $[(x + 2)(x - 2) / (x + 2)(x + 3)] \times [(x + 3) / (x + 2)]$.

Cancel $(x + 2)$ from numerator and denominator of first fraction: $(x - 2)/(x + 3)$.

Multiply by $(x + 3)/(x + 2)$: cancel $(x + 3)$: result = **$(x - 2)/(x + 2)$** .

Q17: Pythagoras + trig

(a) By Pythagoras: $BC^2 = AC^2 - AB^2 = 15^2 - 12^2 = 225 - 144 = 81$.

$BC = \mathbf{9 \text{ cm}}$.

(b) $\cos(\text{BAC}) = \text{adjacent/hypotenuse} = 12/15 = 0.8$.

angle BAC = $\cos^{-1}(0.8) = \mathbf{36.9^\circ}$ (1 d.p.).

Q18: Direct proportion (cube)

(a) $y = kx^3$. When $x = 2$, $y = 56$: $56 = k \times 8$, so $k = 7$.

$$y = \mathbf{7x^3}.$$

(b) $189 = 7x^3 \Rightarrow x^3 = 27 \Rightarrow \mathbf{x = 3}$.

Q19: Histogram

Frequencies = frequency density $\times$ class width:

0–10: $0.8 \times 10 = 8$; 10–20: $2.4 \times 10 = 24$; 20–25: $5.0 \times 5 = 25$;

25–35: $3.0 \times 10 = 30$; 35–50: $1.2 \times 15 = 18$.

(a) Total = $8 + 24 + 25 + 30 + 18 = \mathbf{105 \text{ cars}}$.

(b) Cars exceeding 30 mph: half of 25–35 bar ($5 \text{ wide} \times 3.0 = 15$) + full 35–50 bar (18) = **33 cars**.

Q20: Quadratic sequence

(a) First differences: 6, 10, 14, 18 (increasing by 4).

Second differences: 4, 4, 4. Constant second difference indicates a quadratic sequence.

Coefficient of $n^2 = (\text{second difference}) / 2 = 2$.

Check $2n^2$: 2, 8, 18, 32, 50. Differences from actual: 1, 1, 1, 1, 1.

So nth term = $2n^2 + 1$. ✓

(b) 12th term = $2(12)^2 + 1 = 2 \times 144 + 1 = 289$.

Q21: Transformations

(a) Reflection in $y = x$ swaps x and y coordinates:

$A(1, 1) \rightarrow A'(1, 1)$. $B(3, 1) \rightarrow B'(1, 3)$. $C(3, 2) \rightarrow C'(2, 3)$.

(b) Enlargement scale factor -2 from origin: each coordinate is multiplied by -2 .

$A(1, 1) \rightarrow A'(-2, -2)$.

Q22: Algebraic proof

Let the two consecutive integers be n and $n + 1$.

Difference of squares: $(n + 1)^2 - n^2 = (n^2 + 2n + 1) - n^2 = 2n + 1$.

Sum of the two integers: $n + (n + 1) = 2n + 1$.

Both expressions equal $2n + 1$, so the **difference of squares equals the sum of the integers**. ✓